

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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Management Accounting

CLASS ACTIVITY 6

<https://tinyurl.com/Fall2025-Activity6>

Scenario

A company manufactures three models of smartphones: **Standard, Premium, and Ultra**. Each technician earns an annual salary of **£28,000**, plus the employer contributes **5%** to a pension scheme and pays an **Employer's National Insurance Contribution of 12%**

- Each technician works **1,400 productive hours** per year.
- The company's **annual overhead costs are £50,000**, and it expects to sell **500 Standard units, 300 Premium units, and 200 Ultra units**.

The component and labour requirements are:

- **Standard:** £150 components + 8 hours labour
- **Premium:** £250 components + 10 hours labour
- **Ultra:** £400 components + 14 hours labour

Questions

Question 1: Calculate the **hourly direct labour cost** of a technician.

Question 2: Allocate the **overhead cost equally per unit** (Way 1). What is the total cost per unit for each model?

Question 3: Allocate the **overhead cost proportionally to labour hours** (Way 2). What is the total cost per unit for each model?

Question 4: Allocate the **overhead cost proportionally to total direct cost** (Way 3). What is the new cost per unit for each model?

Values Given

- Annual salary = **£28,000**
- Employer pension = **5%** of salary = **£1,400.00**
- Employer NIC = **12%** of salary = **£3,360.00**
- Total annual employer labour cost = $28,000 + 1,400 + 3,360 =$ **£32,760.00**
- Productive hours per technician per year = **1,400 hours**
- Hourly direct labour cost = $32,760 \div 1,400 =$ **£23.40 / hour**
- Annual overheads = **£50,000**
- Expected sales: Standard = **500**, Premium = **300**, Ultra = **200** (total **1,000** units)
- Components & labour hours per unit:
 - Standard: **£150, 8 hours**
 - Premium: **£250, 10 hours**
 - Ultra: **£400, 14 hours**

Question 1

1) Hourly direct labour cost

Total annual employer cost = 28,000 + 1,400 + 3,360 = **£32,760.00**

Hourly direct labour cost = 32,760 ÷ 1,400 = **£23.40/hour.**

Question 2

Way 1 — Overhead allocated equally per unit

Overhead per unit = $50,000 \div 1,000 = \text{£}50.00$

- Labour cost per unit (direct) = hours $\times$ £23.40
- Standard labour = $8 \times 23.40 = \text{£}187.20$
- Premium labour = $10 \times 23.40 = \text{£}234.00$
- Ultra labour = $14 \times 23.40 = \text{£}327.60$

Total cost per unit = component + labour + overhead:

- **Standard:** $150 + 187.20 + 50.00 = \text{£}387.20$
- **Premium:** $250 + 234.00 + 50.00 = \text{£}534.00$
- **Ultra:** $400 + 327.60 + 50.00 = \text{£}777.60$

Question 3

Way 2 — Overhead allocated proportional to labour hours

- Total labour hours for all units:
- Standard: $500 \times 8 = 4,000$
- Premium: $300 \times 10 = 3,000$
- Ultra: $200 \times 14 = 2,800$
Total = **9,800 hours**

Overhead per labour hour = $50,000 \div 9,800 = \textbf{£5.102041 / hour}$ (approx)

- Loaded hourly labour cost = direct hourly + overhead per hour = $23.40 + 5.102041 = \textbf{£28.502041 / hour}$
- Total cost per unit = component + (labour hours $\times$ loaded hourly cost):
- **Standard:** $150 + 8 \times 28.502041 = 150 + 228.01633 = \textbf{£378.02}$
- **Premium:** $250 + 10 \times 28.502041 = 250 + 285.02041 = \textbf{£535.02}$
- **Ultra:** $400 + 14 \times 28.502041 = 400 + 399.02857 = \textbf{£799.03}$

Question 4

Way 3 — Overhead distributed in proportion to total direct cost

- Direct cost per unit (components + labour at direct hourly £23.40):
- **Standard direct cost:** $150 + 8 \times 23.40 = \text{£}337.20$
- **Premium direct cost:** $250 + 10 \times 23.40 = \text{£}484.00$
- **Ultra direct cost:** $400 + 14 \times 23.40 = \text{£}727.60$

Total direct cost across all units = $500 \times 337.20 + 300 \times 484.00 + 200 \times 727.60 = \text{£}459,320.00$

- Overhead rate = $50,000 \div 459,320 \approx \mathbf{0.1088566}$ ($\approx \mathbf{10.88566\%}$)
- Cost per unit including overhead = direct cost $\times$ (1 + overhead rate):
- **Standard:** $337.20 \times 1.1088566 = \text{£}373.91$
- **Premium:** $484.00 \times 1.1088566 = \text{£}536.69$
- **Ultra:** $727.60 \times 1.1088566 = \text{£}806.80$